Response to the Editor and Reviewer

Manuscript: *A Latent Oscillator Measurement Model to Simulate Emotional-Expression Score Dynamics in Video*

Dear Dr. Epskamp, Editors, and Reviewer,

Thank you for the careful reading and for the guidance on scope, structure, and examples. We revised the paper as a methods contribution centered on the Latent Oscillator Measurement Model (LOMM). The main changes are:

1. *Name and scope.* We renamed the model from DLO-MM to LOMM because damping is one of three velocity-feedback regimes the model allows (damped, undamped, amplifying). This fits short video time series better, because brief expressive episodes in the MAFW clips may show bursts, switching, or low expressivity without clear damping. The paper now has one aim: LOMM generates data with a known factor structure so that analysis methods can be tested for whether they recover that structure.

2. *Structure.* The manuscript has six main sections (Introduction, LOMM, Study 1, Study 2, Results, Discussion) and a pipeline figure (Figure 1). The full model specification is in Appendix A, design and implementation details in Appendix B, and supplementary results in Appendix C (Study 1 diagnostics, grouping and trajectory diagnostics, one-at-a-time sensitivity analyses, the extended factorial simulation, runtime and feasibility, and community-detection sensitivity).

3. *Realism.* A new Study 1 compares simulated score series with classifier scores from 100 MAFW clips on eight summary statistics, using five-fold cross-fitting. MAFW is defined at its first mention in the Introduction.

4. *Benchmarks and metrics.* Study 2 now compares DynEGA with static EGA, GraphicalVAR, and GIMME on identical datasets. It reports recovery with failures and timeouts counted as incorrect, adds assignment accuracy, adjusted Rand index, variation of information, and NRMSE, and reports runtime. Its primary analyses use continuous indicators generated by LOMM; all clips share the same indicator assignments and nominal loadings and residual variances, with small clip-specific perturbations. A separate sensitivity condition applies the softmax directly to the p continuous indicators and analyzes the resulting bounded scores and their centered log-ratio (CLR) components; the original grouping is not preserved, which is why the Study 2 recovery claims are restricted to continuous indicators.

5. *Examples.* The political example is replaced with a non-political MAFW clip.

6. *Tutorial material.* Removed, because the *transforEmotion* tutorial is published separately and is cited in Study 1.

Because line numbers change during production, we refer below to sections, figures, tables, and page numbers in the revised PDF.

Response to the Editor

1. **Editorial guidance on structure and pipeline.**

In your revision, I urge you to especially note the reviewer's comments on restructuring the paper, and also consider including a pipeline figure describing how all components interact.

Location in revised manuscript: Abstract, p. 1; Introduction, pp. 2–3; Figure 1, p. 3; The Latent Oscillator Measurement Model, pp. 3–6; Study 1, pp. 5–8; Study 2, pp. 8–10; Results, pp. 11–15; Discussion, pp. 15–17.

Response: We reorganized the manuscript into six main sections: Introduction, LOMM, Study 1, Study 2, Results, and Discussion. The main text now presents the conceptual model and the decisions needed to understand each study, while exact parameter grids, package settings, and supporting analyses appear in three appendices. Figure 1 gives a two-panel overview. Study 1 compares simulated bounded scores with MAFW-derived classifier scores. Study 2 evaluates recovery from continuous indicators and reports the bounded-score condition as a separate sensitivity analysis.

2. **Editorial guidance on politically sensitive examples.**

In addition, please carefully reconsider the use of politically sensitive examples in the manuscript. My concerns are that (1) the political content may distract readers from the methodological contribution, which should be the focus; (2) the interpretive framing of emotion results (e.g., anger and fear in Boris Johnson's Brexit address) involves editorial judgment that goes beyond what a methods paper typically requires; and (3) politically charged examples risk dating the paper and may be less accessible to an international readership unfamiliar with the specific political context. The methodological points could likely be illustrated equally well with less politically loaded material, which may also be more suitable for guiding empirical researchers aiming to use your method in a lab setting.

Location in revised manuscript: Study 1: Comparing Simulated and Real Score Series, pp. 5–8; Figure 3, p. 7; Discussion, pp. 15–17.

Response: We replaced the political example with a non-political MAFW clip. Figure 3 shows four selected frames from that clip, the corresponding multivariate classifier-score time series, and smoothed classifier-score trajectories. The frames serve only to identify the dataset-derived scoring example. The GitHub repository at `https://github.com/atomashevic/LOMM` does not include raw MAFW videos; access to the original video materials remains governed by the dataset providers. The limitation language states that the softmax-normalized classifier scores are model-derived observations of visible expression, not validated emotion labels or exact empirical trajectories.

Action Editor Comments

3. **Notation for N_g and P_s.**

I find the notation N_g and P_s confusing, especially as N is already used for neutrality, making g and s read as subscripts rather than parts of compound variable names. Please consider clearer notation.

Location in revised manuscript: The Latent Oscillator Measurement Model, pp. 3–6; Study 2: Recovery of Dimensionality and Indicator Assignments, pp. 8–10.

Response: We removed the confusing notation and rewrote the model with explicit indices and dimensions. A notation paragraph in the model section uses $i = 1, \ldots, N$ for clips, $n = 1, \ldots, T$ for observations, $t_n = (n-1)\Delta$ for physical time, and $f = 1, \ldots, m$ for latent dimensions. The

number of bounded score channels is J, their coordinate dimension is $d = J - 1$, and p is the number of continuous indicators; in Study 2, $p = mk$, with k indicators per dimension.

4. **Matrix multiplication notation.**

   Equation (2): The × symbol is used for matrix multiplication; the · symbol or simple juxtaposition would be more conventional in this context.

   Location in revised manuscript: Equations (1) and (2), pp. 4–5; Appendix A, pp. 21–28.

   Response: We rewrote the equations with standard matrix notation. The observation layer in log-ratio coordinates is $\mathbf{u}_{in} = \boldsymbol{\mu} + \mathbf{b}_i + \kappa s_i(\boldsymbol{\Lambda}_i \tilde{\mathbf{x}}_{in} + \boldsymbol{\varepsilon}_{in})$, with bounded outputs $\mathbf{p}_{in} = \text{softmax}(\mathbf{V}\mathbf{u}_{in}/\tau)$. Designs with fixed loadings use $\boldsymbol{\Lambda}_i = \boldsymbol{\Lambda}$ for every clip, and the primary Study 2 design sets $\mathbf{b}_i = \mathbf{0}$ and $s_i = 1$. Appendix A gives the complete specification, including the bounded-score sensitivity condition described in point 4 of the overview above.

5. **Boldfacing of covariance matrices.**

   Line 277: The covariance matrix Q is not boldfaced, inconsistent with the boldfacing convention used for other matrices.

   Location in revised manuscript: Model equations and observation layer, pp. 4–5; Appendix A, pp. 21–28.

   Response: We revised the notation throughout the model section. The current observation layer uses the clip-indexed loading matrix $\boldsymbol{\Lambda}_i$, activity-scaled latent vector $\tilde{\mathbf{x}}_{in}$, measurement-error vector $\boldsymbol{\varepsilon}_{in}$, and covariance matrix $\boldsymbol{\Sigma}_i$. Fixed-measurement designs set $\boldsymbol{\Lambda}_i = \boldsymbol{\Lambda}$ and $\boldsymbol{\Sigma}_i = \boldsymbol{\Sigma}$. Scalar acceleration inputs are written as $q_{if,n}$.

6. **Missing word.**

   Line 304: A word appears to be missing: "We will illustrate this approach [using?] OpenAI's CLIP..."

   Location in revised manuscript: Study 1: Comparing Simulated and Real Score Series, pp. 5–8.

   Response: We removed the old tutorial-style sentence and replaced it with a direct description of the empirical scoring target, the eleven score channels, complete-row retention, the closure check and normalization of small closure drift, and the sensitivity analysis that interpolates internal gaps of at most two frames within a segment. Values are not truncated, and zero replacement is disabled. The empirical scoring section identifies *transforEmotion* as the source of the frame-level softmax-normalized classifier scores, using model key `oai-large`, which denotes the OpenAI CLIP ViT-Large/14 model family.

7. **Interpretive claim in the previous political example.**

   Figure 3 implies that Boris Johnson expressed fear throughout the duration of his address on Brexit. This is a substantial interpretive claim that I do not think is suitable for a methods journal, and it relies on the validity of CLIP, which is not discussed. Furthermore, if this paper is to guide empirical research, it would be more suitable to demonstrate the method using neutral or controlled videos that researchers might encounter in a lab setting.

Location in revised manuscript: Study 1: Comparing Simulated and Real Score Series, pp. 5–8; Figure 3, p. 7; Discussion, pp. 15–17.

Response: We removed the Boris Johnson and Brexit example. Figure 3 uses a non-political MAFW clip and shows selected dataset frames, score series, and smoothed trajectories. The still frames identify the scored clip while the raw MAFW videos remain outside the GitHub repository; access to the original materials remains under the dataset providers' terms. The text avoids substantive political interpretation of a public speech.

8. **CLIP and FER validity.**

> *The method relies heavily on CLIP being able to accurately classify emotions for any given set of emotion labels. However, the manuscript does not cite or discuss evidence for CLIP's validity in facial expression recognition specifically. At minimum, the authors should review existing work on CLIP's FER accuracy and acknowledge limitations.*

Location in revised manuscript: Introduction, pp. 2–3; Study 1: Comparing Simulated and Real Score Series, pp. 5–8; Discussion, pp. 15–17.

Response: We revised the framing so that the score series are treated as softmax-normalized classifier scores rather than validated emotion labels. The Introduction cites work on CLIP-like and zero-shot FER models as scoring context, and the empirical section identifies the *transforEmotion* backend, model key `oai-large`, and OpenAI CLIP ViT-Large/14 model identifier. We address validity by limiting the claim: the score series are prompt-specific classifier scores, and the simulated-versus-real comparison does not validate scoring-model labels, prompt labels, subjective emotional states, or exact empirical trajectories.

9. **Realism of LOMM simulations.**

> *How do we know that DLO-MM produces realistic FER-like data? It would strengthen the paper considerably to compare statistical properties of simulated datasets to those of real CLIP-derived FER data, to demonstrate that the simulation model produces data resembling real-world observations.*

Location in revised manuscript: Study 1: Comparing Simulated and Real Score Series, pp. 5–8; Study 1 Results, pp. 11–12; Figure 4, p. 11; Appendix C.1, pp. 34–40; Discussion, pp. 15–17.

Response: We added a simulated-versus-real comparison using MAFW-derived classifier-score time series. Study 1 evaluates eight summary statistics: median level, interquartile range of the level, interquartile range of frame-to-frame changes, lag-1 autocorrelation, entropy of monotonic run lengths, normalized spectral entropy, effective rank, and total variance. Each statistic is evaluated on the scores and on their log-ratio coordinates, giving 16 comparisons across the retained 100 video clips. The analysis is a five-fold cross-fitted exploratory comparison of eight configurations. The same held-out folds were used to compare the configurations and identify the lowest-discrepancy one, so the comparison is descriptive; an unbiased external assessment would require new data or an outer validation split. Appendix C retains the bootstrap means, estimated biases, basic reverse-percentile intervals, and percentile intervals as diagnostics of the clip-resampling distribution, not as uncertainty intervals around the plug-in discrepancies. Clip baseline produced most of the discrepancy reduction; clip scale contributed less. Study 2 is a separate analysis and is described under the next point.

10. **Alternative dimensionality recovery benchmarks.**

I appreciate the effort in developing a simulation framework, which is not common in image classification work as far as I am aware. However, since the authors develop both the DLO-MM simulation method and the DynEGA analysis method, strong performance of DynEGA on DLO-MM data is not entirely surprising. It would considerably strengthen the contribution to include alternative dimensionality recovery methods as benchmarks, to determine whether DynEGA specifically excels or whether recovering latent dimensions from this type of data is straightforward more generally.

Location in revised manuscript: Study 2: Recovery of Dimensionality and Indicator Assignments, pp. 8–10; Study 2 Results, pp. 12–15; Table 3, p. 12; Figure 5, p. 13; Appendix B, pp. 29–33; Appendix C, pp. 34–52.

Response: We added static EGA, GraphicalVAR, and GIMME to the matched-method simulation design. Static EGA ignores temporal order and is computationally inexpensive; it shows how much of the grouping is recoverable from contemporaneous indicator levels alone. GraphicalVAR is the dynamic network comparison. GIMME is included as a per-clip method whose residual-covariance matrices from valid clip-level fits are averaged, symmetrized, converted to a nonnegative graph by the prespecified rule, and partitioned; this tests computational feasibility as well as recovery.

The main Results report the valid-output rate, the correct-dimension rate among valid outputs, and the overall correct-dimension recovery rate. The last quantity counts every failure and timeout as an unsuccessful recovery. Assignment accuracy is reported among valid network outputs, and runtime and feasibility are counted per unique fitted structure. Appendix C reports the complete grouping diagnostics, trajectory correlation and NRMSE, and all-attempt failure-adjusted trajectory measures; an attempt without a valid network output receives correlation 0 and NRMSE 2.

The comparison shows that recovery was not specific to DynEGA. GraphicalVAR and GIMME are included in the matched-method design, and the comparison applies to the procedures and conditions used there. At $T = 100$, overall correct-dimension recovery was 0.939 for DynEGA, 0.884 for static EGA, 0.777 for GraphicalVAR, and 0.176 for GIMME. Static EGA also exceeded GraphicalVAR in assignment accuracy and required a median 2.15 seconds rather than 85.41 seconds. This result makes clear how much of the target was recoverable from contemporaneous structure. Appendix B documents the exact estimator and graph-conversion procedures, including the GIMME conversion used here.

Response to Reviewer 1

1. **General assessment and scope.**

The paper has the potential to be very timely and interesting to psychologists, as it uses both advanced zero-shot classification (AI-based) and theoretically motivated dynamic modeling (ODE-based). In principle, this is an exciting and valuable direction and I see a lot of applications. I would have been considerably more enthusiastic about the contribution if the manuscript did not suffer from notable lack of coherence, structure, conceptual clarity, and mathematical precision.

Location in revised manuscript: Whole manuscript, especially Abstract, p. 1; Introduction, pp. 2–3; model section, pp. 3–6; Study 1, pp. 5–8; Study 2, pp. 8–10; Results, pp. 11–15.

Response: We narrowed and reorganized the manuscript. LOMM now has one role in the paper: it is a data generator with a known factor structure, used to evaluate whether analysis methods recover the number of dimensions and the indicator assignments.

2. **Introduction and conceptual clarity.**

> *I think the Introduction can be tightened in favor of conceptual clarity, such that the paper quickly comes to the point. Currently, it oscillates between aims, brief snippets of related work, and a restatement of a singular aim (P6L46). I suggest a more brain-friendly structure that briefly introduces emotion recognition, existing difficulties with theory-driven modeling (DLO-MM) and data-driven detection (DL, supervised and zero-shot classification), and how the proposed framework addressed these difficulties. Upon a naive reading, it may currently not become clear whether the authors propose new methods, apply existing methods in a new context, validate their own software, or a hodgepodge of different things revolving around the quantitative analysis of emotion dynamics. Some methodological aspects currently appear out of the blue (e.g., dynEGA) and it is not at all clear which method is being validated for what purpose. For example, only much later does it become clear that DLO-MM will be used as a simulator to validate the accuracy of dynEGA in in silico.*

Location in revised manuscript: Abstract, p. 1; Introduction, pp. 2–3; Figure 1, p. 3.

Response: We rewrote the Introduction to move directly from multivariate classifier-score time series, to the measurement problem, to LOMM, and then to the two studies. Figure 1 appears before the model section and shows the distinct targets. DynEGA, static EGA, GraphicalVAR, and GIMME are introduced together in Study 2 as recovery procedures evaluated against the same prespecified indicator grouping.

3. **Connections among CLIP, simulation, and DynEGA.**

> *Related: It is unclear how the goals of the different methods connect to each other. For instance, even 15 pages deep in the paper, it remains unclear how the goal of generic synthetic emotion scores connects to the goal of applying zero-shot classification to the frames. How exactly do CLIPs outputs feed into DLO-MM or vice versa (or not at all?). Some figures appear at random places (e.g., Figure 3 presages the entire methodology of the paper, even though only DLO-MM has been formally introduced). It would be helpful it the paper included a "pull figure" describing all components of the pipeline and their interrelations (and priming for the tutorial part), instead of relying on readers to piece together the puzzle and fill in the blanks. Additionally, it would be helpful if sections include paragraphs (see AI papers) that group together and delineate core ideas. For instance, CLIP and details about its pretraining appear under "Emotion Dynamics as a Damper Oscillator Model", implying an apparent (but missing) connection.*

Location in revised manuscript: Figure 1, p. 3; Study 1: Comparing Simulated and Real Score Series, pp. 5–8; Study 2: Recovery of Dimensionality and Indicator Assignments, pp. 8–10; Appendix B, pp. 29–33.

Response: Figure 1 shows the full workflow in two panels. In Study 1, empirical softmax-normalized classifier scores provide the comparison target for LOMM bounded scores. In Study 2, LOMM generates continuous indicators with a prespecified grouping for recovery analysis; the bounded-score sensitivity condition is described in point 4 of the overview. The main text keeps these questions separate, and Appendix B gives the design and implementation details.

4. **Uneven detail and tutorial framing.**

Furthermore, the granularity of description of the components of the pipeline differs widely, from more precise and theoretically argued for (DLO-MM), to less detailed with a patchwork of mathematical details and lots of submethods and moving parts (dynEGA), to merely explained on a high level (CLIP, zero-shot classification). Together with the fact that the descriptions of these methods are interleaved throughout the text, but their connections are often left implicit or informal, it becomes incredibly hard to follow the paper, especially as a "tutorial". The formal level of description also experiences an uninterrupted drop from DLO-MM onwards.

Location in revised manuscript: Introduction, pp. 2–3; Study 1, pp. 5–8; Study 2, pp. 8–10.

Response: We moved tutorial framing out of the manuscript because the *transforEmotion* R package tutorial has already been published separately and is cited in Study 1. The main LOMM section now explains the latent, activity, and measurement layers through a running psychometric example, a notation paragraph, two equations, and a worked process figure. Study 1 defines the empirical comparison target, and Study 2 introduces the four recovery procedures through a compact method-role table. Exact mathematical and computational details are collected in Appendices A and B.

5. **Notation and mathematical precision.**

There are some unfortunate choices of notation and mathematical sloppiness. For example, in a dynamic modeling context, Nt (Eq.1) can easily be interpreted as N times t (continuous time). Why do the parameters eta and xi need an Nt subscript? I understand that Eq.1 applies to all three sentiment variables, but positive and negative sentiment are only briefly discussed. Matrices are not consistently bolded (e.g., P13L270). It is unclear where the 2D vector X_t comes from (which is bolded and capitalized as if it were a matrix). It is later redefined as a delay embedding matrix (P20L398). Where does the i index come from (P14L281)? The dynamic errors's symbol is redefined from sigma to q. The time index is redefined from t to i (I assume t was meant to be reserved for continuous time and i for discrete). These are just a few examples of sloppy mathematics, prompting a thorough revision for consistency and precision.

Location in revised manuscript: Model equations, pp. 4–5; observation layer, p. 5; Appendix A, pp. 21–28; outcome measures, Appendix B.4, p. 33.

Response: We rewrote the mathematical exposition and defined each index and object where it first appears. Video is indexed by i, discrete observation by $n = 1, \ldots, T$, physical time is $t_n = (n-1)\Delta$, and latent dimension is indexed by f. The state is $\mathbf{s}_{if,n} = [x_{if,n}, v_{if,n}]^\top$, and the acceleration input is $q_{if,n}$. The current observation layer uses the clip-indexed loading matrix $\mathbf{\Lambda}_i$, measurement error $\boldsymbol{\varepsilon}_{in}$, and covariance matrix $\mathbf{\Sigma}_i$; designs with fixed measurement parameters set the clip-indexed matrices equal across clips.

We also removed the continuous- versus discrete-time ambiguity the reviewer noted. The main text gives the linear second-order oscillator in continuous time and defines the piecewise-constant acceleration input on each observation interval. Appendix A gives the exact sampled transition, $\mathbf{s}_{if,n+1} = \mathbf{\Phi}_f(\Delta)\mathbf{s}_{if,n} + \mathbf{\Gamma}_f(\Delta)q_{if,n}$, including the transition matrix and acceleration-input loading vector. The revised notation omits the old N_t, N_g, and P_s terms.

6. **Data-generation pipeline and relation to CLIP outputs.**

I am left puzzled by some aspects of the data generation pipeline. For example, it is called simulate_video (which would imply that video frames are simulated), but as far as I understood, it simply used DLO-MM to generate score time series. Also a new parameter eta (with no subscript) appears out of nowhere. Further, at a conceptual level, FER scores are said to apply to both sentiments (neutral, positive) and concrete emotions (excitement, fear,...), so in what

sense are the outputs of CLIP compatible, conflicting with, or orthogonal to the outputs of DLO-MM? Can DLO-MM be fitted to CLIP outputs? In the latter case, DLO-MM will be rather a model of AI emotion detection performance than a model of emotion dynamics per se.

Location in revised manuscript: Introduction, pp. 2–3; Study 1, pp. 5–8; Study 2, pp. 8–10; Appendix B, pp. 29–33; Discussion, pp. 15–17.

Response: We revised the terminology so the pipeline no longer implies that video frames are simulated. LOMM generates simulated score series. CLIP outputs do not feed into LOMM and LOMM is not fitted to them; the empirical classifier scores from MAFW serve only as the comparison target in Study 1, where the bounded-score observation layer lets clips differ in baseline level and scale. Study 2 analyzes continuous indicators (overview, point 4); its between-clip-variability analysis draws clip baselines inside the space spanned by the loadings so that the prespecified indicator blocks are preserved. The affect-taxonomy paragraph is explicitly illustrative: approach-affiliation, threat-defense, and orienting-novelty are simulation-design dimensions, and the prompt labels are examples rather than validated affect categories in MAFW.

Appendix B lists the family codes, activity gains, dwell fractions, and numeric η, ζ, and per-step σ_q vectors used in the final runs; η is now defined at first use in the model section. Appendix C reports finite-horizon amplification and stability diagnostics. The Discussion states that the score series are model-derived observations of visible expression, so the current simulation design does not validate emotion labels or claim direct equivalence between empirical trajectories and latent affective states.

7. **Recovery metrics beyond correlation.**

Using correlation for validation the recovery of true scores only tells part of the story. Please, complement the results with a metric that reflects potential biases (e.g., NRMSE, predictive coefficient of determination, ...).

Location in revised manuscript: Study 2, pp. 8–10; Study 2 Results, pp. 12–15; Table 3, p. 12; Appendix C.2, pp. 41–45; Appendix C.5, pp. 49–50.

Response: We added metrics beyond correlation. The primary simulation design reports the overall correct-dimension recovery rate and assignment accuracy, together with runtime among valid network outputs and valid-output, timeout, and error counts. Accuracy is reported per fitted network and community-detection algorithm, while feasibility and runtime are reported per unique fitted structure. The main Results report the $T = 100$ trajectory correlation and NRMSE values and explain that they describe post-estimation recovered-group summaries rather than the primary grouping target. Appendix C gives the complete diagnostics. All-attempt versions assign correlation 0 and NRMSE 2 to attempts without a valid network output.

8. **Calibration and limitations of zero-shot classifiers.**

The calibration of zero-shot classifiers is generally hard to measure or unknown (comapred to the calibration of supervised classifier). That means that CLIP-induced FER scores (which can be interpreted or converted into probabilities) for more exotic emotions (e.g., excitement, pride) have an unknown reliability. The same goes for its accuracy in detecting rare emotions. I believe these points warrant at least a small discussion, as users should know what to expect for their own applications and where to find numbers on accuracy and calibration for common emotion sets.

Location in revised manuscript: Introduction, pp. 2–3; Study 1: Comparing Simulated and Real Score Series, pp. 5–8; Discussion, pp. 15–17.

Response: We describe these outputs as softmax-normalized classifier scores rather than calibrated emotion probabilities. The Introduction cites recent zero-shot and CLIP-like FER work as scoring context. The empirical Methods identify the *transforEmotion* scoring backend. The Limitations section states that the simulated-versus-real comparison evaluates statistical signatures of score series rather than the validity of emotion labels.

9. **Acronyms and model name.**

   Some acronyms are not introduced, redundantly introduced multiple times, or introduced following a different term (e.g., DLO is hidden among references on P7L52, but the term is Damped Oscillator Model, which would abbreviate to DOM).

   Location in revised manuscript: Title and Abstract, p. 1; Introduction, pp. 2–3; model section, pp. 3–6.

   Response: We changed the model name to Latent Oscillator Measurement Model (LOMM) and introduce it directly in the main title, abstract, and Introduction. Damping is one optional velocity-feedback regime in the model. The model section now explains damped, undamped, and amplifying velocity-feedback cases.

10. **Neutral factor.**

    Why is there no neutral factor in the measurement model? The latent neutral trajectory is plotted in Figure 2 but its relationship to the observable scores is unclear.

    Location in revised manuscript: Observation layer, p. 5; Appendix A, pp. 21–28; Figure 2, p. 6; Study 1, pp. 5–8.

    Response: In Study 1, neutral is one of the eleven score channels; like every channel, its log-ratio coordinate is a linear function of the m latent dimensions plus clip baseline and measurement error, so it is not a separate factor. In Study 2 the indicators are generic and carry no emotion labels. The new Figure 2 shows a generic latent process, activity schedule, expressed signal, and four fallible indicators, and no longer suggests a separate neutral trajectory.

11. **Equation 1.**

    Eq. 1 needs attention. The first term literally evaluates to $dN_t/dt = dN_t/dt$ and is completely redundant. I suspect the zero and the one in the "parameter matrix" have to be swapped.

    Location in revised manuscript: Equation (1), p. 4; Appendix A, pp. 21–28.

    Response: We replaced the old equation with a standard second-order state representation. The first row now maps velocity to the derivative of position, and the second row maps position feedback, velocity feedback, and the acceleration input to acceleration.

12. **Dynamic error.**

    It is not clear where the "dynamic error" enters the model and what makes it dynamic (it is not introduced as a function of time). Since this is a tutorial, I would have expected a more precise and focused mathematical exposition.

Location in revised manuscript: Equation (1), p. 4; Appendix A, pp. 21–28; Figure 2 and accompanying text, pp. 4–6.

Response: We now define the acceleration input as $q_{if,n} \sim \mathcal{N}(0, \sigma^2_{qf})$, held constant within an observation interval and entering the exact state transition at each step. The main LOMM text explains its role in creating irregular peaks and reversals. Appendix Figure A1 shows trajectories with and without these inputs.

13. **Idiosyncratic expressions.**

> *Nit: Some expressions are rather idiosyncratic, e.g., "Emotion Dynamics as a Damped Oscillator Model" -> the notion of "X as a model" seems strange, a more natural expressions would be "Modeling emotion dynamics via ...", "quantitative methods approach" is redundant -> perhaps "a quantitative framework / method / approach", there are lots of synonyms.*

Location in revised manuscript: Title, p. 1; The Latent Oscillator Measurement Model, pp. 3–6; manuscript-wide prose revisions.

Response: We revised the title and section titles to better match the paper's methodological contribution. We also reduced tutorial-style and redundant phrasing throughout the revised manuscript.

14. **Figure captions.**

> *Figure captions are sloppy, use inconsistent capitalization, and rarely helpful or self-explanatory.*

Location in revised manuscript: Figure 1, p. 3; Figure 2, p. 6; Figure 3, p. 7; Figure 4, p. 11; Figure 5, p. 13; Appendix Figures C1–C6, pp. 35–52.

Response: We rewrote the figure captions to be self-contained and consistent. The five main figures now provide a two-study overview, a four-stage LOMM process, the empirical score example without a single-clip network panel, the Study 1 discrepancy on the score representation, and Study 2 recovery across scenarios at $T = 100$ and $T = 25$. Supporting sensitivity, runtime, and community-detection figures appear in Appendix C.

Closing

We appreciate the editor's and reviewer's guidance. The revision specifies LOMM as a data generator with an explicit measurement model, compares selected features of simulated score series with MAFW-derived classifier-score time series, and compares dimensional-recovery methods against a prespecified factor structure while counting failures and timeouts. The main recovery claims concern continuous indicators; a bounded-score sensitivity condition shows that their original grouping is not preserved after direct softmax normalization, whether the transformed data are analyzed as raw bounded scores or CLR components. Although the empirical illustration uses video-derived classifier scores, the revision makes explicit that LOMM's methodological contribution is the separate specification of latent dynamics, activity, and observation for simulation-based evaluation under a prespecified recovery target. We hope the revised manuscript makes the paper's scope, evidence, and limits easier to evaluate.

Sincerely,

The Authors